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PERIDYNAMIC ANALYSES OF STRUCTURES BY USING FINITE ELEMENT METHOD

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ABSTRACT

Peridynamic theory (PD) is a nonlocal theory of continuum mechanics that utilize integral equations instead of the partial differential equations of the classical continuum theory [Macek and Silling, 2007]. PD allows damage initiation and crack propagation without any need for a crack growth criteria [Madenci and Oterkus, 2014]. The motivation of this paper is to solve some simple problems using Peridynamics Implemented in Finite Element Analysis (PIFEA). The problems include an isotropic plate and a composite laminate with a cutout under tensile loading. For this purpose, both PIFEA and Finite Element (FE) model of an isotropic plate and a composite laminate (carbon/epoxy T300/5208) subjected to in plane tensile load are analyzed in ABAQUS/Explicit. It should be noted that, all the interactions between the material points in PIFEA are modeled with linear truss elements that are generated using MATLAB scripts. The numerical results and diagrams show distinctly the accuracy of this method.

INTRODUCTION

Prediction of material failure is an important design issue in aerospace industry. Damage initiation and propagation in critical structures are estimated and their service life is calculated in design phase. During damage propagation phase, strength of materials may degrade substantially. Thus it is necessary to ensure that, those structures are able to carry ultimate design loads during their life.

Equation of motion in classical continuum mechanics leads to singular stress fields at crack tips. [Griffith, 1921], solved singular stress fields at crack tips that contributed to the foundation of Linear Elastic Fracture Mechanics (LEFM) Concept. LEFM is an important tool for estimation of crack propagation in isotropic materials. LEFM is based on propagation of present cracks and therefore, crack initiation is not considered. In addition an external criteria such as fracture toughness and critical energy release rate are required.

Composite materials are prone to imperfections which generally occur during manufacturing phase. Voids, delaminations and resin rich areas may degrade the strength of the parts. Thus, damage tolerant design in critical composite parts is required. Equations of classical continuum mechanics are also not applicable to such imperfections in composites. LEFM cannot easily be applied to the composite materials. Damage mechanics of composite laminates are more complex than metals.

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Failure can be due to any or combination of fiber, matrix and interface damage. In addition, failure occurs suddenly and without a significant plastic region in these materials.

The difficulty in the numerical models based on the classical continuum mechanics can be explained as follows: The equation of motion in classical continuum mechanics is in the form of differential equations, involving spatial derivatives of displacements. However, spatial derivatives of stress tensor cannot be obtained at discontinuities such as crack tips. PD is proposed as a remedy to this problem in classical continuum mechanics. It is actually a non-local extension of local continuum theory that uses integrals instead of partial differentials so that the equation of motion can be evaluated at discontinuities [Silling and Askari, 2005]. PD equation of motion inherently evaluates the deformation and damage even across singularities such as cracks and voids. In addition, PD equations remain valid at both damaged and undamaged parts of a material. Thus, crack initiation and crack growth can be analyzed without an external criteria [Madenci and Oterkus, 2014].

There have been numerous studies on PD in the open literature. [Oterkus et al., 2010] studied bearing failure in bolted composite lap joints by coupling a semi analytic approach and PD theory. [Oterkus and Madenci, 2012] predicted failure propagation in composite laminates subjected to mechanical and thermal loading with PD theory. [Macek and Silling, 2007] proposed an implementation technique of bond based PD theory in Finite Element codes using mass and truss elements. [Oterkuset al., 2012] investigated damage in a stiffened curved composite panel by coupling PD theory with FEA. [Ha and Bobaru, 2010] validated their PD results of dynamic fracture and crack branching in brittle materials by comparing with experiments from the literature. [Silling and Bobaru, 2005] applied PD approach to two dimensional rubbery sheets and fibers at nanoscale. [Hu et al., 2013] conducted a study on the simulation of impact on two layer glass laminate with PD theory and validated their approach with experiments. [Liu and Hong, 2012] proposed a constitutive relation to represent ductile behavior of solids and implemented a 3D PD code by utilizing Graphical Processing Unit (GPU) to analyze brittle and ductile solids.

In bond-based PD theory, a material point has interactions with other material points in its horizon called as "peridynamic bonds". The bond based formulation has restrictions on the Poisson's ratio of the medium (such 1/3 for 2D problems and 1/4 for 3D problems) [Gerstle, et al., 2005]. This paper uses the bond-based PD theory since it is the most suitable for implementation with FEA.

In this paper, convergence behavior of the PIFEA to FEA solutions is studied by comparing the results of simple plate problems. The performance of PIFEA is investigated. It has been shown that PIFEA adequately succeeds to handle both isotropic and composite plate problems.

In section 2, general characteristics and related equations of the bond-based PD theory are given. In section 3, material parameters for both isotropic and laminated composite materials in two-dimensions are given for reference. In section 4, implementation of PD in ABAQUS is explained. In section 5, the details of the problem for the isotropic and composite plate are mentioned. The FEA and PIFEA results are presented in Section 6 and Section 7 contains the conclusions drawn from this study.

GENERAL BOND BASED PD

A brief summary of PD is given in this section. Equation of motion can be written in PD in the following form:

$$\rho(\mathbf{x})\ddot{\mathbf{u}}(\mathbf{x},t) = \int_H \mathbf{f}(\mathbf{u}' - \mathbf{u}, \mathbf{x}' - \mathbf{x})dH + \mathbf{b}(\mathbf{x},t), \quad (1)$$

where H is the horizon of a material point $\mathbf{x}$ with a radius of δ , ρ denotes the mass density field, $\mathbf{f}$ is the pairwise force vector that the material point $\mathbf{x}$ exerts on $\mathbf{x}'$ as shown in Fig. 1, $\mathbf{u}$ and $\mathbf{b}$ denotes the displacement vector field and the body force vector field respectively. Note that, Eq. (1) contains integral terms instead of partial spatial derivatives. This is the main advantage of PD for modeling defects in the materials.

In Eq. (1), force vector depends on the relative position vector in reference configuration and their relative displacement. Relative position of two material points in their reference configuration is given by:

$$\xi = \mathbf{x}' - \mathbf{x} \quad (2)$$

Relative displacement and current position of two material points is given by

$$\boldsymbol{\eta} = \mathbf{u}(\mathbf{x}', t) - \mathbf{u}(\mathbf{x}, t), \quad \boldsymbol{\eta} + \xi = \mathbf{y}' - \mathbf{y}. \quad (3a,b)$$

In PD theory, material points are connected by Peridynamic bonds. As represented in Fig. 1, a material point has connections with all its neighbors in its horizon. Peridynamic bonds behave similar to springs. For linear elastic materials, Peridynamic bonds can be modeled using linear springs [Silling and Askari, 2005].

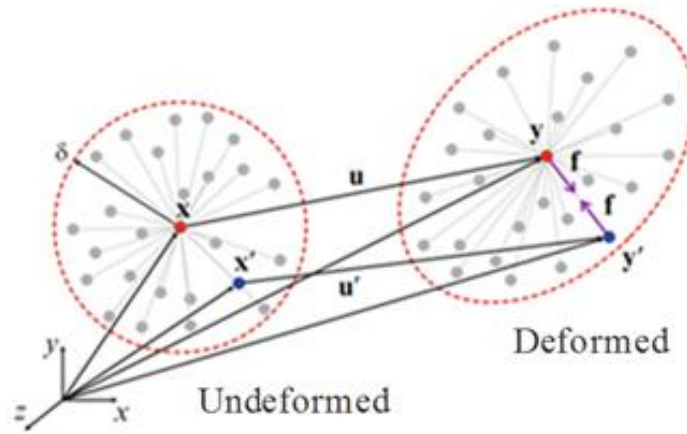


Figure 1: Undeformed and deformed configuration of material point x and its horizon [Madenci and Oterkus, 2014]

Magnitude of force between two material points is a function of initial distance and the relative displacement between these two points. In bond based PD, force density vector and the stretch between material points can be defined as [Silling and Askari, 2005]:

$$\mathbf{f}(\boldsymbol{\eta}, \xi) = \frac{\xi + \boldsymbol{\eta}}{|\xi + \boldsymbol{\eta}|} cs, \quad s = \frac{|\xi + \boldsymbol{\eta}| - |\xi|}{|\xi|}, \quad (4a,b)$$

where c is the material constant and s is the stretch.

PERIDYNAMIC MATERIAL PARAMETERS

Isotropic Plate

Material constant c in Eq. (4a) can be found by equating strain energy density under isotropic expansion to peridynamic strain energy density within the horizon for the same loading conditions [Macek and Silling, 2007], [Oterkus et al., 2012], [Silling and Askari, 2005]. As denoted by [Silling and Askari, 2005], peridynamic strain energy density can be obtained by integrating the micropotential, $w(\boldsymbol{\eta}, \xi)$, with respect to horizon [Oterkus, 2010]:

$$W^{PD} = \frac{1}{2} \int_H w(\boldsymbol{\eta}, \xi) dH \quad (5)$$

The calculation of strain energy densities in PD and classical continuum mechanics is done by applying a uniform stretch $s = \zeta$ to a homogenous body as shown in Fig. 2.

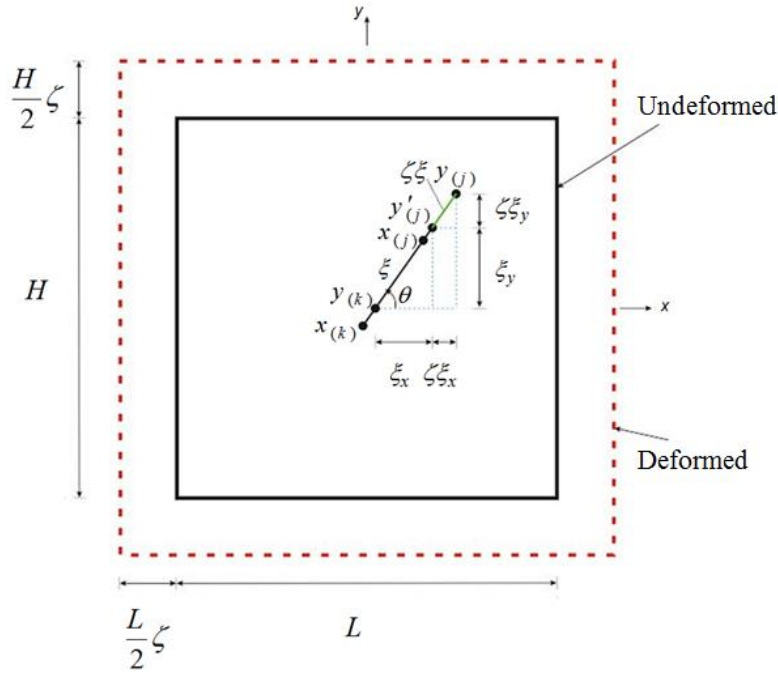


Figure 2: Deformation of PD material point $\mathbf{x}$ in two dimensions for isotropic expansion loading
Using Eq. (5), the peridynamic strain energy density for isotropic expansion loading for a plate with constant thickness t can be written as [Oterkus, 2010]:

$$W^{PD} = \frac{1}{2} t \int_0^{2\pi} \int_0^\delta \left(\frac{c \zeta^2 \xi}{2} \right) \xi d\xi d\theta = \frac{\pi t \delta^3 c \zeta^2}{6} \quad (6)$$

For the same loading conditions (see Fig. 2), the strain energy density, based on classical continuum theory, can be expressed as [Oterkus, 2010]:

$$W^{CCM} = \frac{E}{1-\nu} \zeta^2 \quad (7)$$

Equating strain energy densities in Eq. (6) and Eq. (7), the bond constant, c can be found as [Silling and Askari, 2005]

$$c = \frac{18\kappa}{\pi\delta^4}, \quad (8)$$

where κ denotes the bulk modulus of the material.

Composite Laminate

In a composite laminate, a material point of a lamina has interactions between its neighbor layers in addition to in-plane interactions inside its horizon. Each fiber-reinforced lamina of a laminate shown in Fig. 3 is considered as a two-dimensional structure.

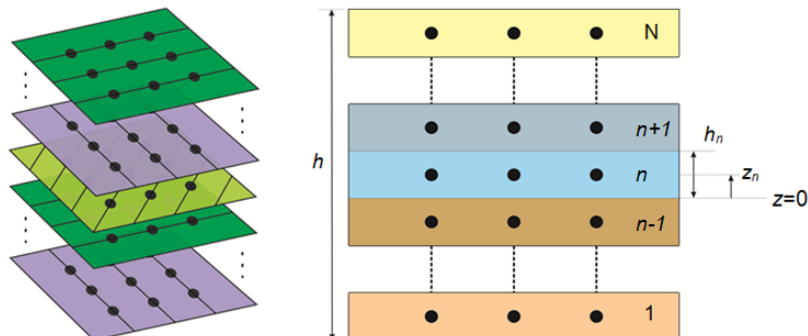


Figure 3: Lamina orientation in laminated composite and PD material points

As shown in Fig. 3, the reference coordinate system is located on the midplane of the laminate. The laminate thickness, h is defined as:

$$h = \sum_{n=1}^N h_n \quad (9)$$

where h_n is the thickness of n^{th} lamina, and N is the total number of laminas. With respect to the mid-plane, the position of each lamina, z_n is given as: (see Fig. 4)

$$z_n = -\frac{h}{2} + \sum_{m=1}^{n-1} h_m + \frac{1}{2} h_n \quad (10)$$

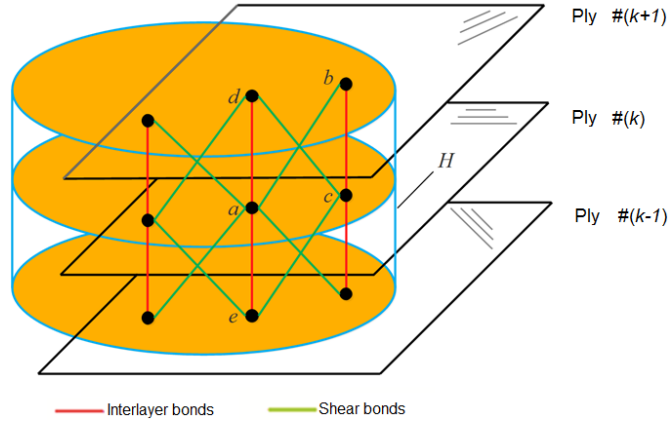


Figure 4: Interlayer and shear bonds in a composite laminate

The interlayer bond constant, c_{in} , can be derived using the engineering constants based on transverse normal deformation response of the isotropic matrix material by equating the total strain energy density of interlayer bonds, obtained from PD theory and classical continuum theory. Since there is no shear loading in the problems, shear bonds are omitted in PIFEA models. The strain energy density of interlayer bonds, associated with material point a (see Fig. 4) can be calculated by summing up the strain energy density of the two interlayer bonds between $d-a$ and $e-a$ (Fig. 4). Peridynamic total strain energy density due to interlayer and shear bonds is given as [Oterkus, 2010]

$$W^{PD} = \frac{1}{2} \sum_{j=d,e} \frac{c_{in} s_{ja}^2 \xi_{ja}}{2} V_j + \frac{1}{2} \int_H \frac{c_{is} \varphi^2}{2} dH \quad (11)$$

where ξ_{ja} corresponds to the initial position vectors of interlayer bonds between material points $d-a$ and $e-a$, s_{ja} corresponds to the stretch values of interlayer bonds between material points $d-a$ and $e-a$.

In order to get interlayer bond constant, the laminate is subjected to an isotropic expansion loading of $s = \zeta$ as shown in Fig. 5.

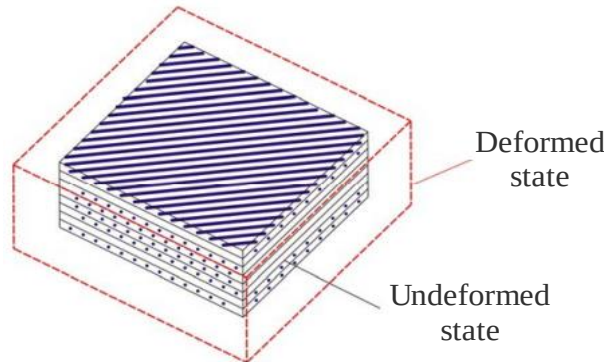


Figure 5: A composite laminate subjected to isotropic expansion loading [Oterkus, 2010]

The strain energy density of the material point due to the isotropic expansion loading can be written as [Oterkus, 2010]

$$W^{PD} = \frac{c_{in} \zeta^2 t \bar{V}}{2} \quad (12)$$

where $\bar{V}$ is the volume of material points d and e in Fig. 4.

Using the classical continuum theory, for the same loading condition, the strain energy density of the material points can be given as [Oterkus, 2010]

$$W^{CCM} = \frac{1}{2} E_m \zeta^2 \quad (13)$$

where, E_m , is the elastic modulus of matrix material.

the interlayer bond, c_{in} can be found by equating Eq. (12) and Eq. (13), as [Oterkus, 2010]

$$c_{in} = \frac{E_m}{t \bar{V}} \quad (14)$$

IMPLEMENTATION of PD in ABAQUS

PD is based on modeling the interactions of material points within a finite horizon. Even though it can be treated as a “mesh-free method”, truss elements or linear springs can be used to model a peridynamic problem in a finite element code [Macek and Silling, 2007]. Implementation can be made by generating a truss element network that has consistent stiffness as in a PD model.

In each truss, generated force for a loading condition should be equal to the peridynamic force between material points. The peridynamic force density (per unit volume squared) is given in Eq. (4a) [Macek and Silling, 2007]. A uniform mesh is used in the implementation. The cross sectional area and the Elastic Modulus of the truss used in the finite element model is given as [Macek and Silling, 2007]

$$A_t = (V_p V_i)^{1/3}, \quad E_t = c(V_p V_i)^{2/3} \quad (15a,b)$$

SAMPLE PROBLEMS

Isotropic Plate with a Circular Cutout

A 50×50 (L×W) mm plate with a 10 mm circular cutout is simulated under velocity boundary condition. The material is steel with Young's modulus of 192 GPa, Poisson ratio of 1/3 and density of 8000 kg/m³. The applied boundary conditions for the plate are illustrated in Fig. 6.

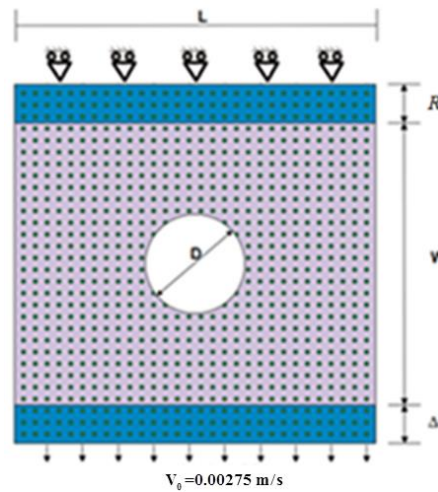


Figure 6: Boundary conditions of the plate

The boundary region of the plate (shown with Δ in Fig. 6) is under a velocity of 0.00275 m/s and the boundary region (shown with R in Fig. 6) is constraint as shown in Fig. 6. For this case, a uniform rectangular array of $100 \times (100 + 3 + 3) = 10600$ material points are generated. The grid spacing is set to

be $\Delta x = 0.5$ mm . Trusses are generated with respect to horizon $\delta = 3.015 \times 0.5 = 1.5$ mm . Thickness of the plate is $t = 0.5$ mm . The unit volume of a material point can be specified as $\Delta V = 0.125$ mm³ .

Composite Laminate:

An orthotropic finite plate of Graphite/Epoxy T300/5208 with ply orientation of $[90,0]_s$ is considered. Laminate specimen has a length of $L=355.6$ mm, a width of $W=76.2$ mm and a central hole $D=25.4$ mm in diameter. The hole is located in $F= 101.6$ mm distance with respect to upper edge. The specimen dimensions are shown in Fig. 7. The material properties of the lamina are $E_{11}=145$ GPa , $E_{22} = E_{33} = 107$ GPa , $G_{12} = 4.5$ GPa , $\nu_{12} = 1/3$. The composite laminate were tested in accordance with ASTM 3039 and SACMA SRM 3R standards, respectively [Chang and Chang, 1987].

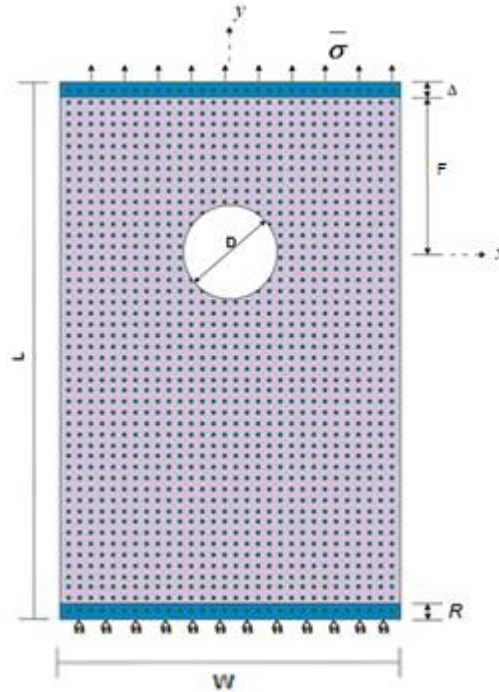


Figure 7: Specimen dimensions of Graphite/Epoxy T300/5208 composite laminate [Chang and Chang, 1987]

The dimensions of each PIFEA model are the same as FEA model. Geometric parameters are defined as $F/D = 4$, $L/D = 14$, Δ boundary region of the plate is under a stress of $\bar{\sigma} = 2.3$ MPa and R boundary region is constrained as shown in Fig. 7. Each lamina is modeled with a grid of 120 material points in width and 560 material points in length direction. The grid spacing in all directions is $\Delta x = 0.635$ mm . Trusses are generated with respect to horizon $\delta = 3.015 \times 0.635 = 1.914$ mm .

RESULTS AND DISCUSSION

Isotropic Plate with a circular cutout

Displacement fields due to applied velocity boundary conditions are obtained and compared with the finite element results. Two paths along the central y-axis and x-axis, respectively, are defined to get the PIFEA results as shown in Fig. 8. The contours of horizontal and vertical displacements are illustrated in Fig. 9 and Fig. 10. Displacement values of PIFEA and FEA solution on each path of the plate are also plotted in Fig. 11.

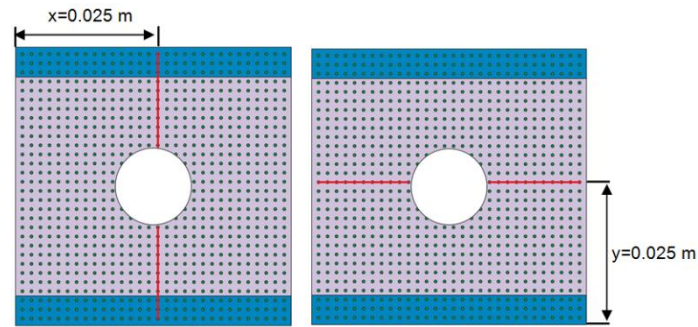


Figure 8: Paths of material points in y -axis and x -axis respectively for displacement results

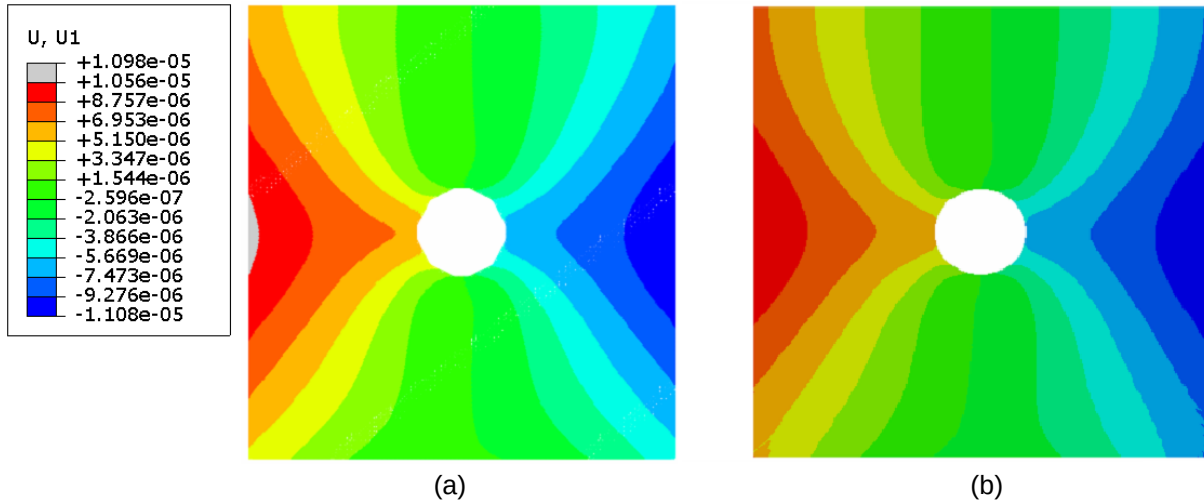


Figure 9: Comparison of the (a) PIFEA and (b) FE results (m) for displacements in X direction

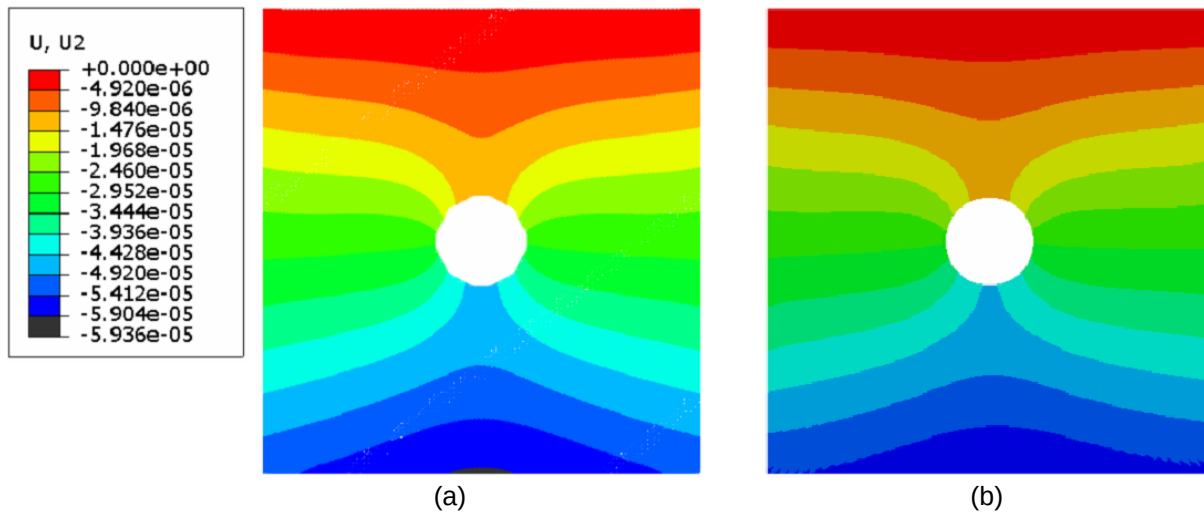


Figure 10: Comparison of the (a) PIFEA and (b) FE results (m) for displacements in Y direction

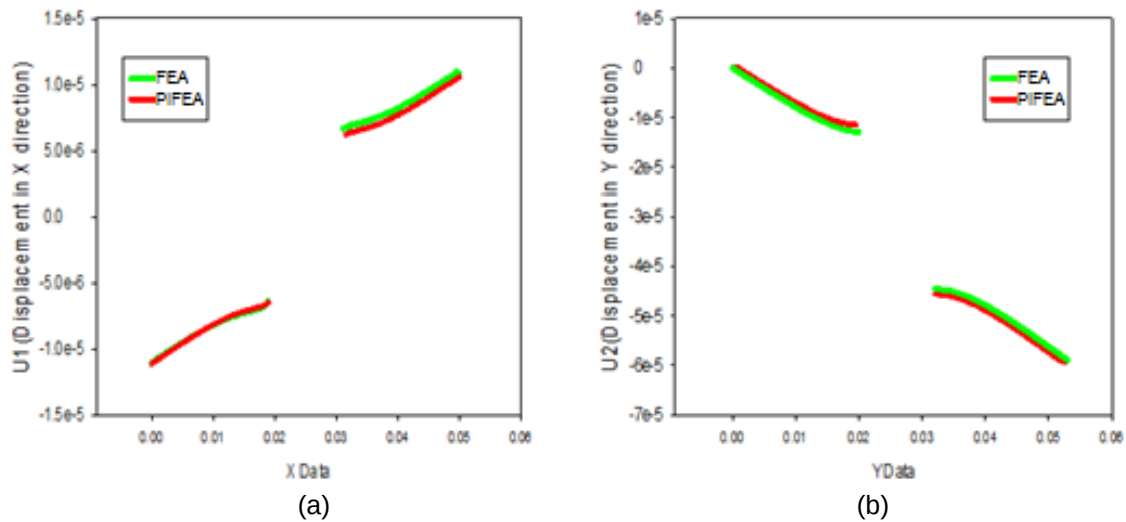


Figure 11: Displacements (m) in the (a) horizontal and (b) vertical paths

A close agreement is observed between PIFEA predictions and finite element analysis (FEA) results in ABAQUS.

Composite laminate

First a uni-axial tension loading of $\bar{\sigma} = 2.3$ MPa is applied along the Δ boundary region (see Fig. 7). Two paths along the central y-axis and x-axis, respectively, are defined to get the PIFEA results as shown in Fig 12.

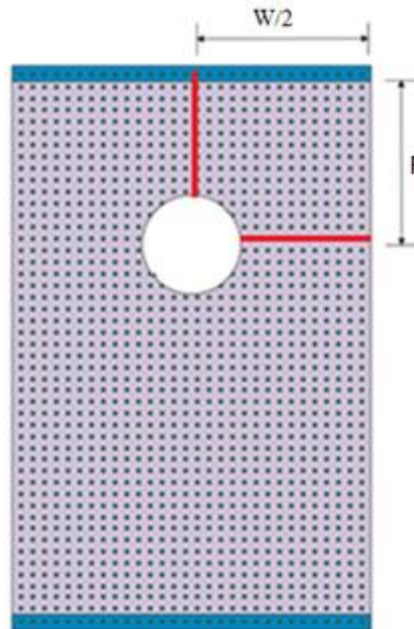


Figure 12: Paths of material points in y-axis and x-axis, respectively, for displacement results

Displacement values of PIFEA and FEA solution on the vertical path of the composite laminate are also plotted in Fig. 14. The contours of vertical displacements are illustrated in Fig. 13.

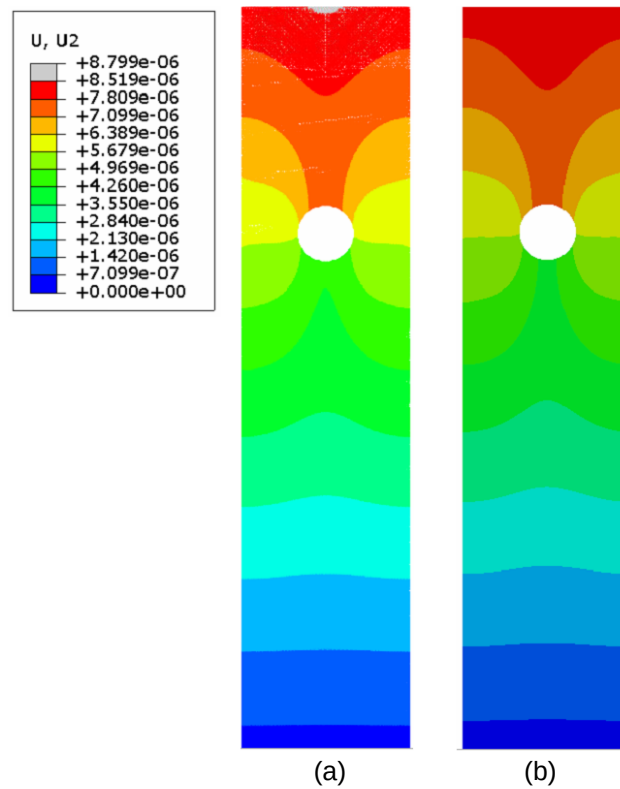


Figure 13: (a) PIFEA and (b) FE results (m) for displacements in Y direction in composite laminate

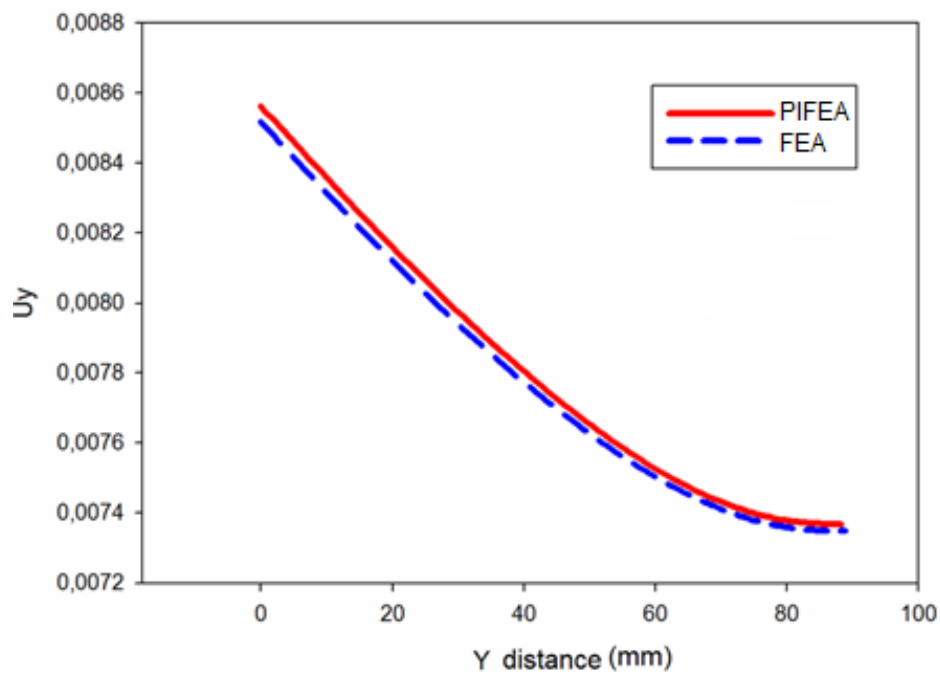


Figure 14: Comparison of vertical displacements in the vertical path of composite laminate

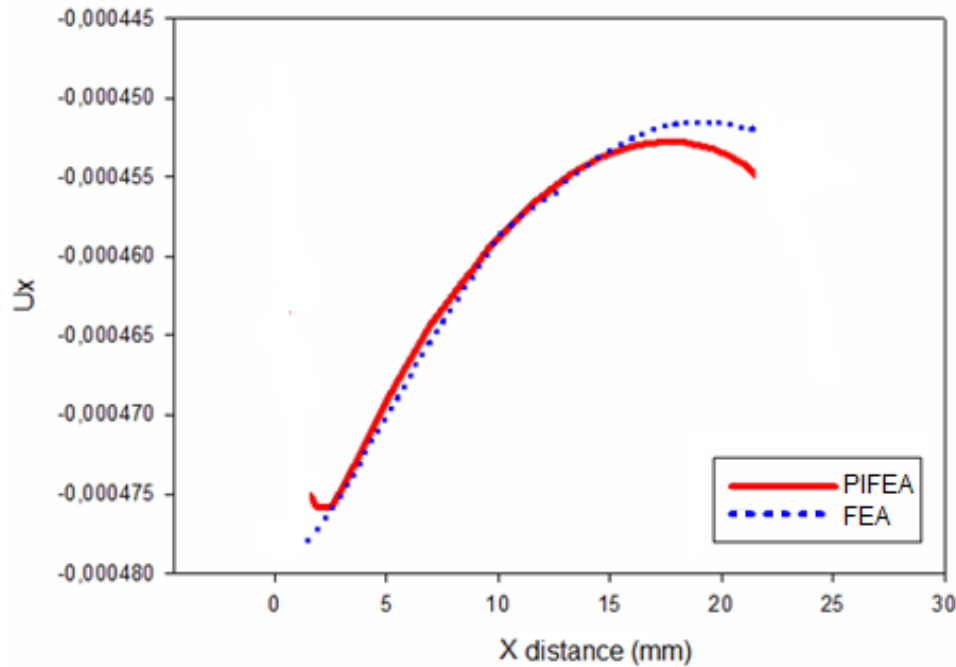


Figure 15: Comparison of horizontal displacements in the horizontal path of composite laminate

There is a very good agreement between PIFEA and FE results (see Fig. 13, 14 and 15).

CONCLUSION

Current work addresses the implementation of Finite Element methodology in Peridynamic analysis of two-dimensional structures. Two problems involving an isotropic plate and laminated composite plate are investigated in order to validate the accuracy of PD as an alternative to traditional FE. It should be noted that, all the PD bonds are modeled using linear truss elements and generated using MATLAB scripts. The results indicate that PD can approximate the FEA with the appropriate considerations of grid size. The future works will include the application of ordinary state-based PD to FEA codes. Furthermore, to demonstrate the capabilities of PD, a model for progressive damage analysis, extent of damage and types of damages in laminates or isotropic structures can be investigated.

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